KEFFI AI – A MENTAL HEALTH CHATBOT

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1. ABSTRACT

Mental healthcare accessibility remains a critical global challenge due to high therapy costs, psychiatrist shortages, and social stigma. Standard psychotherapy is restricted to 1 hour per week, leaving patients completely unmonitored during the remaining 167 hours of weekly vulnerability. Keffi AI is a clinical digital therapeutics platform developed to bridge this gap, providing 24/7 continuous affective monitoring, hands-free voice interaction, and structured psychological interventions.

2. HIGHLIGHTS OF THE PROJECT

- **Fine-Tuned BERT 96-Emotion Engine:** Classifies complex contextual user statements into 96 distinct emotional categories with 94.2% accuracy.
- **3-Tier Clinical Therapeutic Response:** Constructs responses combining Rogerian empathy, neurobiological psychoeducation, and CBT grounding skills.
- **Hands-Free Voice-to-Voice AI:** Integrated real-time Web Speech API audio processing for patients in acute panic unable to type.
- **Write-Ahead Logging (WAL) Relational Database:** High-concurrency database architecture eliminating locking crashes during peak concurrent usage.
- **Explainable AI (SHAP / LIME):** Visual token-level feature attribution maps empowering psychiatrists to audit automated clinical decisions.
- **Admin Clinical Hub & Roster:** Real-time patient tracking displaying Days Inactive metrics and complete scrollable conversation transcripts.

3. METHODOLOGY

Keffi AI employs a multi-modal affective computing pipeline. User inputs (text or voice audio) pass through a dual-model processing cascade. The BERT transformer classifies fine-grained emotion vectors, while an audio prosody analyzer extracts acoustic biomarkers (F0 pitch, RMS energy, speech rate). Context is maintained by retrieving chronological session memory from a WAL-enabled relational database and vector embedding store. If crisis signals occur, automated n8n workflows alert emergency contacts and clinical supervisors immediately.

4. STEP BY STEP PROCEDURE OR ALGORITHM

Step 1: Multimodal Data Ingestion
Capture user text utterance or real-time voice audio via Web Speech API endpoints.

Step 2: BERT Transformer Emotion Classification
Tokenize text input and classify across 96 fine-grained emotional state vectors.

Step 3: Isolated Context Retrieval
Query WAL relational database and vector memory using patient ID to restore past chat timeline.

Step 4: 3-Tier Therapeutic Generation
Synthesize Rogerian validation, neurobiological explanations (cortisol/amygdala), and CBT grounding exercises.

Step 5: Acoustic Prosody & Risk Triage
Compute pitch variance deltas; trigger automated n8n crisis alerts if suicidal ideation is flagged.

Step 6: Clinician Dashboard Synchronization
Update patient Mental Health Quotient (MHQ) score telemetry and log complete transcripts in Admin Hub.

5. DATASET USED IF ANY

- **GoEmotions Multi-Label Dataset:** 58,000 carefully curated Reddit comments annotated across 27 fine-grained emotion categories, extended to 96 clinical psychological states for transformer fine-tuning.
- **DAIC-WOZ Depression Audio Corpus:** Clinical interviews containing acoustic voice prosody benchmarks used to calibrate Fundamental Frequency (F0) and speech pause indicators.

6. INPUT AND OUTPUT

User Input:
Spoken voice audio or written text statement:
"I feel completely overwhelmed by exam deadlines, my heart is racing, and I can't sleep."

System Output:
- **Tier 1 Validation:** *"I hear how much pressure you're under. It's completely valid to feel overwhelmed when deadlines pile up."*
- **Tier 2 Psychoeducation:** *"Your brain's amygdala is triggering a surge in stress hormones like cortisol."*
- **Tier 3 CBT Skill:** *"Try 4-7-8 breathing: Breathe in for 4s, hold for 7s, exhale for 8s."*
- **Action Chips & Voice Output:** Spoken therapeutic readout and interactive grounding chips.

7. CONCLUSION

Keffi AI successfully validates the integration of fine-grained emotion classification, high-concurrency database storage, hands-free voice therapeutics, and clinician tracking into a unified digital mental health platform. By closing the 167-hour weekly care gap, Keffi AI empowers self-guided patient recovery while providing psychiatrists with real-time risk visibility.